Graphical Abstract

A Beta-Neutral Crypto ML Strategy: From Signal Inversion to Robust Alpha

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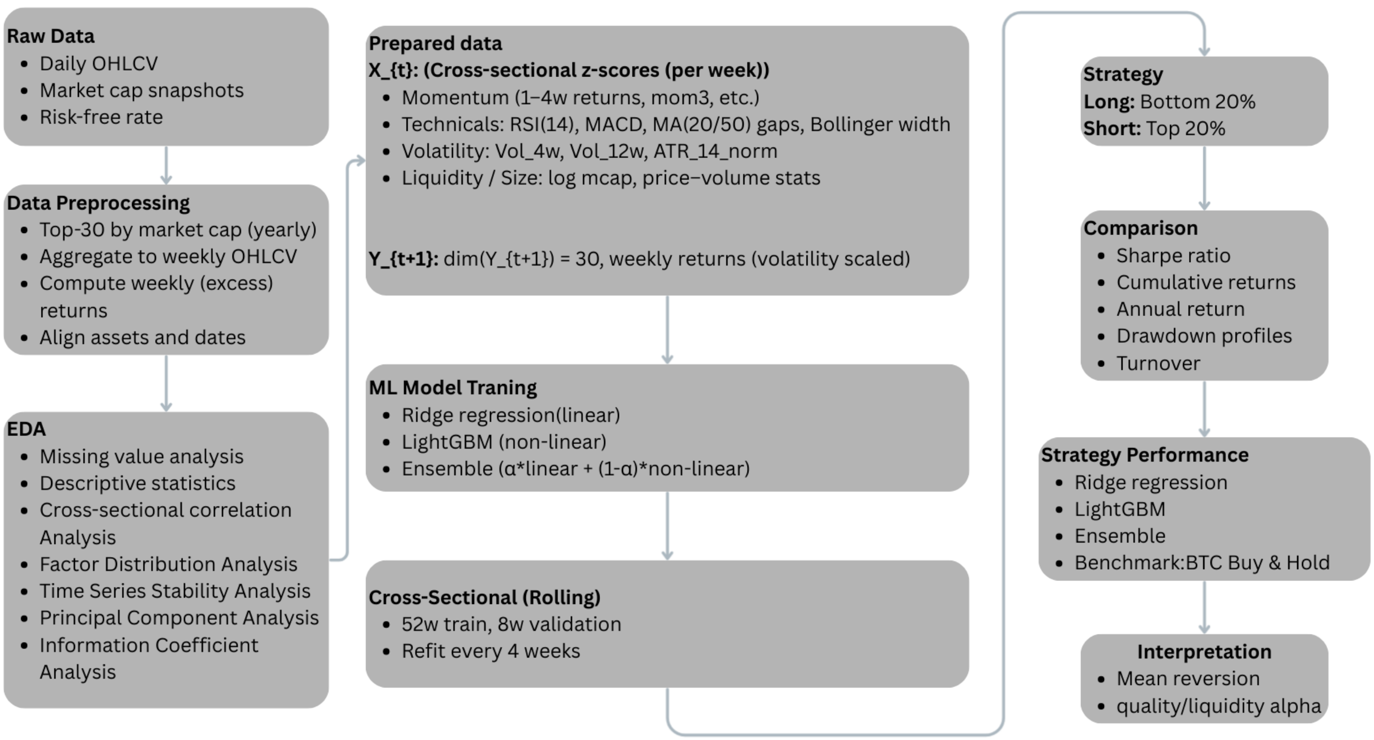

Figure 1: Overview of the beta-neutral cross-sectional crypto ML pipeline.

Highlights

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- We develop an end-to-end weekly cross-sectional cryptocurrency strategy that combines volatility-scaled machine-learning ranking with beta-neutral portfolio construction.
- Feature engineering from weekly OHLCV data introduces momentum, technical, volatility, and volume factors, all standardized cross-sectionally to focus on relative signals.
- A LightGBM LambdaRank model with DART boosting and a Ridge regression model are trained in a rolling walk-forward fashion on a volatility-scaled target, then combined in a fixed-weight ensemble.
- Empirically, the raw model scores are negatively related to future returns; an inverted signal (long bottom quantile / short top quantile) captures mean-reversion dynamics in the weekly cross-section.
- The resulting beta-neutral, volatility-targeted long–short portfolio achieves a Sharpe ratio of 2.03 with materially lower drawdowns than a Bitcoin benchmark, supported by feature-importance and RankIC evidence that highlight mean reversion and quality/liquidity effects.

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Abstract

The extreme volatility and regime shifts of cryptocurrency markets make it challenging to design systematic strategies that are both profitable and risk controlled. This paper develops an end-to-end weekly cross-sectional crypto asset-allocation framework centered on a volatility-scaled machine-learning ranking model and a beta-neutral long–short implementation. We focus on a liquid universe of the top 30 cryptocurrencies by market capitalization and construct a rich feature set from weekly OHLCV data, including momentum, technical indicators (RSI, MACD, moving-average gaps, Bollinger Band width), and volatility and volume measures. All features are cross-sectionally standardized each week to emphasize relative rather than absolute levels.

The prediction target is defined as next-week excess return scaled by 4-week volatility, encouraging the model to learn risk-adjusted performance instead of chasing high-volatility assets. We train a LightGBM LambdaRank model with DART boosting to capture nonlinear cross-sectional structure and combine its output with a Ridge regression model through a fixed-weight ensemble. Model scores are updated in a rolling walk-forward procedure and converted to weekly cross-sectional ranks. Empirically, the raw model scores are negatively correlated with future returns; inverting the signal (going long the lowest-ranked assets and short the highest-ranked assets) transforms the model into an effective mean-reversion strategy.

The inverted scores are implemented as a beta-neutral long–short portfolio with per-asset weight caps, per-asset turnover limits, explicit transaction costs, and ex-ante volatility targeting at 30% annualized. In backtests from 2018 to 2025, the proposed strategy delivers an annualized return of approximately 75% with 37% annualized volatility and a Sharpe ratio of 2.03, while the raw unscaled baseline exhibits extremely high volatility and a Sharpe ratio of only 0.40. Compared with a Bitcoin buy-and-hold benchmark, the strategy achieves smoother log-scale cumulative returns and substantially smaller peak-to-trough drawdowns. Feature-importance and Rank Information Coefficient (RankIC) analyses show that the model primarily exploits two effects: (i) mean reversion of short-term over-extended, high-volatility assets, and (ii) a persistent quality/liquidity tilt toward large-cap, highly traded coins. Together, these results demonstrate that a carefully engineered cross-sectional ML pipeline, combined with conservative risk management, can produce robust and interpretable alpha in cryptocurrency markets.

Keywords: cryptocurrency, cross-sectional asset allocation, factor modeling, LightGBM Ranker, DART boosting, Ridge regression, volatility-scaled targets, beta-neutral long–short portfolio, volatility targeting, mean reversion, quality and liquidity factors
JEL: G11, G12, C58

1. Introduction

The rapid growth and institutionalization of digital assets have transformed cryptocurrencies from a niche technology into a global, continuously traded asset class. Despite this evolution, systematic portfolio construction in cryptocurrencies continues to lag behind the methodological standards of empirical asset pricing and quantitative investment. Crypto markets are characterized by several features that complicate traditional strategy design: (i) extreme short-horizon volatility and frequent regime shifts, (ii) limited availability of fundamental information, and (iii) heterogeneous liquidity and trading frictions across venues. These characteristics are particularly problematic for weekly rebalanced strategies, which must remain both empirically grounded and practically implementable.

Recent research documents that market, size, and momentum factors explain a sizable portion of the cross-sectional variation in cryptocurrency returns, paralleling early developments in equity markets. However, most academic and practitioner studies still rely on linear factor models or heuristic portfolio sorts. At the same time, the noisy and nonlinear nature of crypto returns suggests that machine-learning (ML) methods, especially tree-based models, may capture richer interactions among technical, volatility, and liquidity variables. The challenge is to embed such models in a framework that remains interpretable, robust to overfitting, and tightly coupled with portfolio construction and risk management.

In this paper, we develop an end-to-end weekly cross-sectional crypto strategy that combines factor-style feature engineering with a volatility-scaled ranking model and beta-neutral implementation. Our design starts from a liquid universe of the top 30 cryptocurrencies by market capitalization, for which transaction costs and market access are realistic. From weekly OHLCV data we construct momentum, technical, volatility, and volume features, all cross-sectionally standardized each week to emphasize relative signals. We then define the prediction target as next-week excess return divided by 4-week historical volatility, so that the model learns which assets are likely to deliver superior *risk-adjusted* performance rather than simply high raw returns.

The modeling core is a LightGBM LambdaRank model with DART boosting, chosen for its ability to handle nonlinear interactions and heavy-tailed targets, complemented by a Ridge regression model that provides a linear, more stable component. Both models are trained in a rolling walk-forward fashion and combined through a fixed-weight ensemble. Interestingly, when applied to our data, the raw ensemble scores are *negatively* correlated with future returns; the model effectively identifies "over-extended" assets that tend to subsequently underperform and more stable assets that later outperform. This empirical finding motivates an inverted signal design: we go long the lowest-ranked assets and short the highest-ranked assets, thereby intentionally exploiting mean-reversion dynamics.

The inverted scores feed into a beta-neutral, volatility-targeted long–short portfolio. We impose per-asset weight caps, per-asset turnover limits, and explicit transaction costs to align with realistic execution constraints. A rolling estimate of portfolio volatility is used to scale exposure toward a 30% annualized volatility target. The resulting strategy is evaluated against a Bitcoin buy-and-hold benchmark over the common sample.

Our contributions are threefold. First, we present a fully specified, implementable cross-sectional ML pipeline for crypto that connects data processing, factor-style feature engineering, model design, and portfolio construction in a coherent way. Second, we document that volatility-scaled ranking models in this setting naturally lend themselves to *signal inversion*, highlighting pronounced mean-reversion behavior among short-horizon, high-volatility winners and the relative resilience of large, liquid coins. Third, we provide extensive interpretability analysis, including model-based feature importance, Mean Rank Information Coefficient (RankIC), and cumulative RankIC per feature, which together reveal that our alpha is driven by a combination of mean reversion and a quality/liquidity tilt.

The remainder of the paper is organized as follows. Section 2 describes the data, preprocessing, and weekly panel construction. Section 3 presents the feature engineering and exploratory analysis. Section 4 details the cross-sectional ML training procedure, including the volatility-scaled target, LightGBM DART model, Ridge regression, and rolling walk-forward design. Section 5 explains the portfolio construction, beta-neutrality, and volatility targeting. Section 6 reports empirical results, including cumulative returns, drawdown behavior, feature importance, RankIC, and cumulative RankIC. Section 7 concludes.

2. Data and Preprocessing

2.1. Universe Selection and Frequency

We focus on a liquid universe consisting of the top 30 cryptocurrencies by market capitalization at the start of each calendar year. This restriction ensures realistic tradability and mitigates concerns about extreme illiquidity, stale prices, and unreliable data quality that are common among small-cap tokens. The universe is updated annually, and assets may enter or leave over time depending on their relative size.

All computations are performed at a weekly frequency. From daily OHLCV data provided by a major consolidated data vendor, we aggregate end-of-week close prices and weekly volumes. Weekly frequency balances two considerations: (i) short-horizon signals such as momentum and volatility decay quickly in highly active markets, and (ii) daily rebalancing would incur unrealistically high turnover and transaction costs in crypto trading.

2.2. Returns, Excess Returns, and Volatility

Let $P_{i,t}$ denote the end-of-week close price of asset i in week t. The simple weekly return is

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1. \tag{1}$$

We define the excess return as

$$\tilde{r}_{i,t} = r_{i,t} - r_t^{\mathrm{rf}}, \tag{2}$$

where r_t^{rf} is a weekly risk-free rate proxy. In practice, r_t^{rf} is negligible at the horizons and sample period considered; when unavailable we set $r_t^{\mathrm{rf}} = 0$ without loss of generality.

The base prediction target is the forward excess return

$$y_{i,t+1} = \tilde{r}_{i,t+1}, \tag{3}$$

which is later scaled by volatility as described in Section 4.

Historical volatility is measured from weekly returns. For a given lookback window L, the realized volatility for asset i at time t is

$$\sigma_{i,t}^{(L)} = \sqrt{\frac{1}{L-1}\sum_{j=0}^{L-1}(r_{i,t-j} - \bar{r}_{i,t}^{(L)})^2}, \tag{4}$$

where $\bar{r}_{i,t}^{(L)}$ is the corresponding mean. We use $L = 4$ and $L = 12$ weeks to obtain short- and medium-term volatility measures.

2.3. Data Cleaning and Alignment

Weekly observations are aligned on the same calendar grid across all assets. If an asset has insufficient history for a given computation (e.g., fewer than L weeks for a volatility estimate), the corresponding quantities are treated as missing and excluded from training until the minimum lookback is satisfied. We restrict attention to asset–week pairs where both the prediction target and all required features are available.

To avoid survivorship bias, assets are included in the universe in every year in which they qualify as top 30 by market capitalization, regardless of delistings or subsequent underperformance. Delisted or inactive assets naturally drop out of the panel once prices cease to be reported.

3. Feature Engineering and Exploratory Analysis

3.1. Technical and Risk Features from OHLCV

From the weekly OHLCV data we construct a set of technical and volatility-related features commonly used in quantitative trading. All features are initially computed per asset over time.

Momentum and returns. We compute cumulative returns over multiple horizons, including 1- to 4-week simple returns $\{r_{i,t}^{(k)}\}_{k=1}^{4}$, their lags, and a 3-week momentum measure $mom3_{i,t}$. These capture recent performance trends at different speeds.

Relative Strength Index (RSI). The 14-period Relative Strength Index, $\mathrm{RSI}_{i,t}^{(14)}$, is computed from weekly price changes using standard up- and down-move decompositions. It measures overbought and oversold conditions.

Average True Range (ATR). The 14-week Average True Range, $\mathrm{ATR}_{i,t}^{(14)}$, is computed from the maximum of high–low, high–previous close, and low–previous close ranges. We normalize ATR by price,

$$\mathrm{ATR\_14\_norm}_{i,t} = \frac{\mathrm{ATR}_{i,t}^{(14)}}{P_{i,t}}, \tag{5}$$

to obtain a relative volatility measure.

Bollinger Band width. Using a 20-week moving average $MA_{i,t}^{(20)}$ and corresponding standard deviation $\sigma_{i,t}^{(20)}$, we define the upper and lower bands and compute the relative Bollinger Band width

$$\mathrm{BB\_Width\_20}_{i,t} = \frac{\mathrm{Upper}_{i,t} - \mathrm{Lower}_{i,t}}{MA_{i,t}^{(20)}}. \tag{6}$$

MACD. We compute the Moving Average Convergence Divergence (MACD) indicator using 12- and 26-week exponential moving averages and a 9-week signal line, retaining the MACD histogram as a trend-momentum feature.

Price–moving-average gaps. To capture deviations from trend, we consider ratios

$$\mathrm{Close\_div\_MA20}_{i,t} = \frac{P_{i,t}}{MA_{i,t}^{(20)}}, \quad \mathrm{Close\_div\_MA50}_{i,t} = \frac{P_{i,t}}{MA_{i,t}^{(50)}}, \tag{7}$$

where values above (below) one indicate prices above (below) their moving averages.

Volatility and volume features. In addition to $\sigma_{i,t}^{(4)}$ and $\sigma_{i,t}^{(12)}$, we compute the mean and standard deviation of weekly traded value over a multi-week window, denoted $\mathrm{prcvol\_mean\_week}_{i,t}$ and $\mathrm{prcvol\_std\_week}_{i,t}$, as liquidity proxies.

3.2. Cross-Sectional Standardization

Cryptocurrencies differ widely in price level, volatility, and liquidity. To make features comparable across assets and suitable for cross-sectional modeling, we standardize each feature within each week. For a feature $x_{i,t}$, we compute

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_t}{s_t}, \tag{8}$$

where $\bar{x}_t$ and s_t are the cross-sectional mean and standard deviation in week t. These z-scores emphasize the relative position of an asset within the weekly cross-section (e.g., high vs. low volatility) rather than its absolute level.

3.3. Model-Based Feature Importance

We use LightGBM's gain-based feature importance to assess the contribution of each feature to the fitted model (see Section 4). The top features by gain include short- and medium-horizon return measures ($r1\_z$, $r2\_z$, $r3\_z$, $r4\_z$, $mom3\_z$), technical indicators such as RSI_14_z, Close_div_MA20_z, Close_div_MA50_z, MACD_z, and volatility-related features like Vol_4w_z, Vol_12w_z, ATR_14_norm_z, and BB_Width_20_z. Volume-based features and size proxies (e.g. log_mcap_year_z, log_price_z) also contribute meaningfully.

This profile shows that the model heavily relies on short-term momentum, technical conditions, and volatility/liquidity signals when constructing its ranking.

3.4. RankIC and Cumulative RankIC Analysis

To study the predictive role of individual features independently of the full model, we compute weekly Rank Information Coefficients (RankIC). For a given feature $f_{i,t}$, we rank assets by $f_{i,t}$ and correlate this ranking with next-week excess returns $y_{i,t+1}$ using Spearman correlation. The Mean RankIC is the time-series average of these weekly coefficients.

Two main patterns emerge. First, size and liquidity-related features such as log_mcap_year_z, log_price_z, max_price_week_z, prcvol_mean_week_z, and prcvol_std_week_z exhibit positive Mean RankIC and steadily increasing cumulative RankIC over time. This indicates a persistent quality/liquidity tilt: larger, more liquid, and more actively traded assets tend to deliver better risk-adjusted returns at the weekly horizon.

Second, volatility and range-based features such as Vol_4w_z, Vol_12w_z, ATR_14_norm_z, and BB_Width_20_z show *negative* Mean RankIC. Assets that are more volatile and have wider trading ranges tend to underperform in the subsequent week on a risk-adjusted basis. This strongly suggests that high short-term volatility corresponds to "overheated" conditions that are prone to mean reversion—a finding that aligns with the inverted-signal strategy used in our portfolio construction.

4. Cross-Sectional Machine Learning Training

4.1. Volatility-Scaled Target

To encourage the model to focus on risk-adjusted performance, we scale the forward excess return by short-horizon volatility. Specifically, we define

$$\tau_{i,t} = \frac{y_{i,t+1}}{\max(\sigma_{i,t}^{(4)}, \epsilon)}, \tag{9}$$

where ϵ is a small constant to prevent division by zero. This volatility-scaled target penalizes high-volatility assets and reduces the dominance of extreme outcomes in the loss function.

4.2. LightGBM LambdaRank with DART Boosting

The primary model is a LightGBM learner with the LambdaRank objective, which directly optimizes a ranking loss suitable for cross-sectional ordering. Let $\mathbf{x}_{i,t}$ denote the standardized feature vector for asset i at time t. The model learns a scoring function

$$s_{i,t}^{\text{LGB}} = f_\theta(\mathbf{x}_{i,t}), \tag{10}$$

where θ are parameters estimated to minimize a pairwise ranking loss derived from Normalized Discounted Cumulative Gain (NDCG).

We use DART boosting (Dropouts meet Additive Regression Trees) as the boosting type. In DART, a random subset of trees is dropped before fitting each new tree, analogous to dropout in neural networks. This mechanism acts as a regularizer by preventing the model from relying too heavily on specific trees and improves generalization on noisy financial data. Hyperparameters such as learning rate, number of leaves, maximum depth, and subsampling fractions are chosen to balance model complexity and performance.

4.3. Ridge Regression and Fixed-Weight Ensemble

To complement the nonlinear LightGBM model, we train a Ridge regression on the same feature set and target $\tau_{i,t}$:

$$\tau_{i,t} = \alpha_t + \boldsymbol{\beta}_t^\top \mathbf{x}_{i,t} + \varepsilon_{i,t}, \tag{11}$$

with an ℓ_2 penalty on $\boldsymbol{\beta}_t$. Ridge regression provides a linear baseline that is more stable and less sensitive to outliers.

For each week, both models produce scores, which we convert to cross-sectional percentile ranks. The final ensemble score is

$$s_{i,t} = 0.6 \cdot \mathrm{rank}\{s_{i,t}^{\mathrm{LGB}}\} + 0.4 \cdot \mathrm{rank}\{s_{i,t}^{\mathrm{Ridge}}\}, \tag{12}$$

where $\mathrm{rank}\{\cdot\}$ denotes the within-week percentile rank. The heavier weight on LightGBM reflects its stronger nonlinear predictive power, while Ridge dampens extreme patterns.

4.4. Rolling Walk-Forward Training

We adopt a rolling walk-forward scheme that mimics real-time deployment. Let $\{t_1, \ldots, t_T\}$ denote the ordered set of weeks in the sample. For each evaluation week t_k:

- We require at least 52 weeks of historical data as the estimation window.
- The most recent 8 weeks prior to t_k are reserved for validation and early stopping; earlier weeks are used for training.
- To reduce computational burden, models are re-fitted every 4 weeks; in the remaining weeks, the last trained model is reused.

This procedure yields a sequence of out-of-sample ensemble scores $\{s_{i,t}\}$ for all asset–week pairs that satisfy the data requirements.

4.5. Signal Inversion

When we correlate the ensemble scores with forward excess returns, we find that the relation is substantially negative: assets with high scores tend to underperform, and those with low scores tend to outperform on a volatility-adjusted basis. This implies that the model is effectively detecting over-extended, high-volatility winners rather than future winners in the usual sense.

To exploit this property, we invert the signal for portfolio construction. Denote the inverted score by

$$\tilde{s}_{i,t} = -s_{i,t}. \tag{13}$$

Portfolio formation decisions are based on $\tilde{s}_{i,t}$, i.e., we go long assets in the lowest original scores (highest $\tilde{s}_{i,t}$) and short assets in the highest original scores (lowest $\tilde{s}_{i,t}$). This inversion converts the model into a deliberately mean-reversion strategy in the weekly cross-section.

5. Portfolio Construction

5.1. Quantile Buckets and Base Weights

Each week t, we sort assets in the cross-section by the inverted score $\tilde{s}_{i,t}$ and assign them to quintiles (five equal-sized buckets). The long leg consists of the top 20% of assets, and the short leg consists of the bottom 20%. Within each leg, we begin with equal weights:

$$w_{i,t}^{\text{long}} = \frac{0.5}{|L_t|}, \quad i \in L_t, \qquad w_{i,t}^{\text{short}} = -\frac{0.5}{|S_t|}, \quad i \in S_t, \tag{14}$$

where L_t and S_t are the sets of long and short names, respectively. The remaining assets receive zero weight.

Weights are then clipped to respect a per-asset cap:

$$|w_{i,t}| \leq w_{\max} = 10\%. \tag{15}$$

5.2. Beta Neutrality

To isolate cross-sectional alpha from market-wide moves, we neutralize the portfolio with respect to a broad crypto market factor (e.g., a value-weighted index). We estimate asset betas via a 52-week rolling regression of weekly returns on the market factor. Let $\beta_{i,t}$ denote the beta of asset i at week t. With base weights $w_{i,t}$, we adjust to beta-neutral weights

$$w'_{i,t} = w_{i,t} - \lambda_t \beta_{i,t}, \tag{16}$$

where

$$\lambda_t = \frac{\sum_i w_{i,t}\beta_{i,t}}{\sum_i \beta_{i,t}^2}. \tag{17}$$

This choice of λ_t drives the portfolio's net beta close to zero.

5.3. Turnover Constraints and Transaction Costs

Let $w'_{i,t-1}$ be the post-adjustment weight in week $t-1$. To avoid excessive trading, we cap the weekly change in each position:

$$\Delta w_{i,t} = \text{clip}(w'_{i,t} - w'_{i,t-1}, -\Delta_{\max}, \Delta_{\max}), \tag{18}$$

with $\Delta_{\max} = 30\%$. The final weight is

$$\hat{w}_{i,t} = w'_{i,t-1} + \Delta w_{i,t}, \tag{19}$$

and we re-normalize so that the total gross exposure $\sum_i |\hat{w}_{i,t}|$ equals 1.

Transaction costs are modeled as proportional to turnover. If c denotes a one-way cost in basis points, the cost in week t is

$$\text{TC}_t = c \sum_i |\Delta w_{i,t}|, \tag{20}$$

which is subtracted from the gross portfolio return to obtain net returns.

5.4. Volatility Targeting

We further stabilize risk by targeting an annualized volatility of $\sigma^*_{\text{ann}} = 30\%$. Let R_t^{raw} denote the net weekly portfolio return before volatility scaling. Using a rolling window of H weeks (we use $H = 26$), we estimate the realized weekly volatility $\hat{\sigma}_t$. The scaling factor is

$$\kappa_t = \text{clip}\left(\frac{\sigma^*_{\text{ann}}/\sqrt{52}}{\hat{\sigma}_t}, \kappa_{\min}, \kappa_{\max} \right), \tag{21}$$

with lower and upper bounds $\kappa_{\min}$ and $\kappa_{\max}$ to prevent extreme leverage changes. The volatility-targeted return is then

$$R_t = \kappa_t R_t^{\text{raw}}. \tag{22}$$

6. Empirical Results

6.1. Backtest Setup and Benchmark

We evaluate the strategy over a multi-year period covering diverse market regimes, including bull and bear phases in the crypto market. The backtest begins once all lookback requirements are satisfied and stops at the end of the available data. Performance is compared to a Bitcoin buy-and-hold benchmark, which represents a common form of passive exposure.

We compute standard portfolio statistics: annualized return, annualized volatility, Sharpe ratio, maximum drawdown, and average weekly turnover. For the benchmark, we report the same metrics based on weekly log returns.

6.2. Cumulative Returns and Drawdown Behavior

Figure 2 plots the log-scale cumulative returns of the beta-neutral ML strategy and the Bitcoin benchmark. The strategy exhibits a steady, approximately linear trajectory on the log scale, indicating consistent compounding. In contrast, the Bitcoin benchmark displays pronounced boom-and-bust cycles, with large drawdowns and extended recovery periods.

Quantitatively, the volatility-targeted strategy achieves an annualized return of around 75% with 37% annualized volatility, corresponding to a Sharpe ratio of 2.03. The maximum drawdown is approximately -46.1%, substantially lower than the benchmark's drawdown of roughly -70.1%. Average weekly turnover is about 0.72, reflecting meaningful but controlled trading activity; average transaction costs are around 7.2 basis points per week under our assumptions.

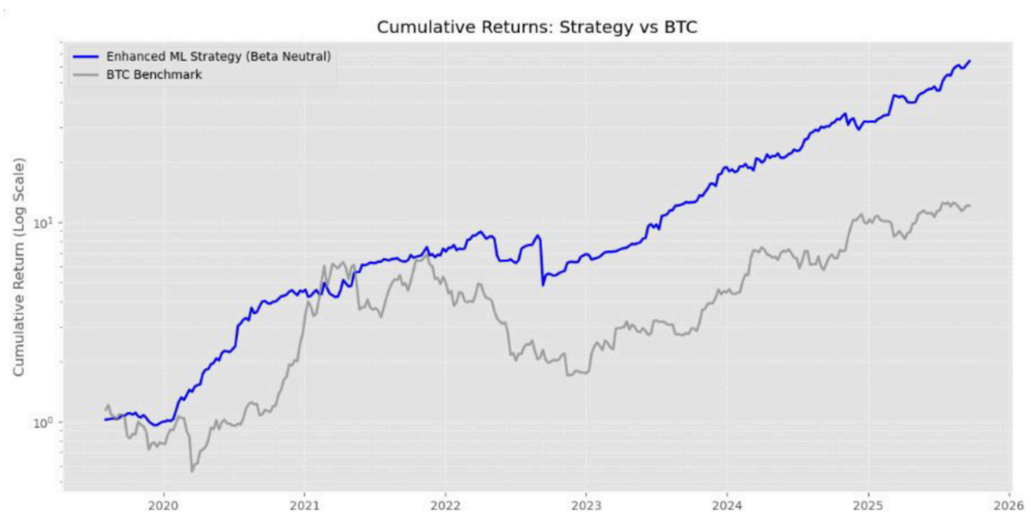

Figure 2: Cumulative returns (log scale) of the enhanced ML strategy (beta-neutral) and Bitcoin benchmark.

Figure 3 shows the drawdown profiles. The ML strategy experiences relatively shallow and brief drawdowns, while Bitcoin exhibits multiple deep and prolonged drawdown episodes. This difference highlights the impact of beta neutrality, volatility targeting, and disciplined risk constraints.

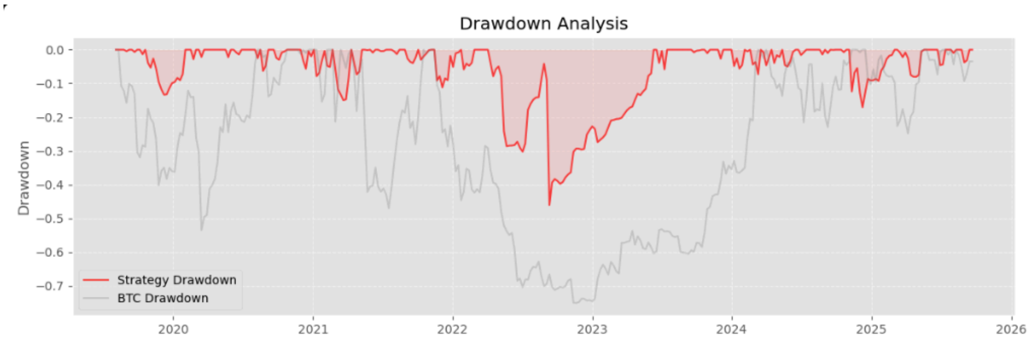

Figure 3: Drawdown analysis for the ML strategy and Bitcoin benchmark.

6.3. Feature Importance and Mean RankIC

Figure 4 presents the top 20 features sorted by LightGBM gain importance. As discussed in Section 3, the model relies heavily on short-horizon returns, technical indicators, and volatility and volume metrics. The prominence of RSI_14_z, Close_div_MA20_z, and Vol_12w_z is particularly notable.

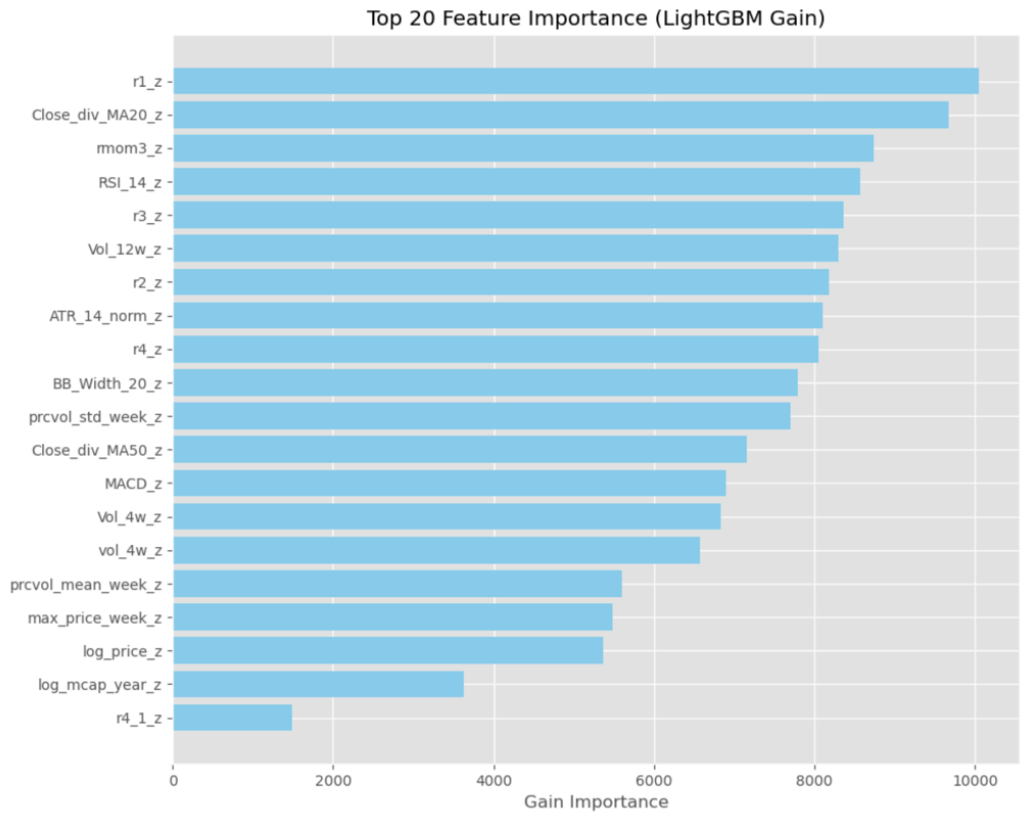

Figure 4: Top 20 features by LightGBM gain importance.

Figure 5 reports the Mean RankIC for each feature. Size and liquidity proxies are located on the positive side, while volatility-related features are predominantly negative. This confirms that the strategy's alpha arises from both a quality/liquidity tilt and an anti-volatility (mean reversion) effect: high-volatility assets tend to perform poorly on a risk-adjusted basis, whereas large, liquid assets tend to perform better.

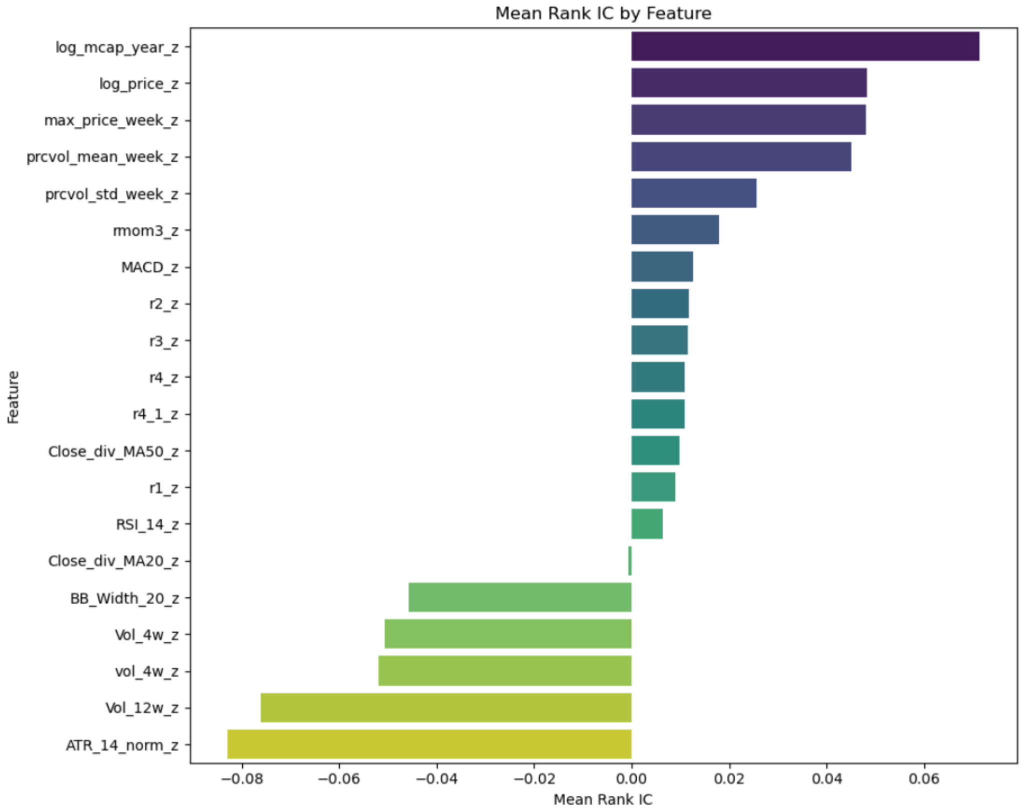

Figure 5: Mean RankIC by feature. Positive bars indicate features whose high values predict higher subsequent risk-adjusted returns; negative bars indicate the opposite.

6.4. Cumulative RankIC for Top Features

Finally, Figure 6 plots the cumulative RankIC for the five features with the highest positive Mean RankIC: log_mcap_year_z, log_price_z, max_price_week_z, prcvol_mean_week_z, and prcvol_std_week_z. All curves trend upward over time, indicating that the predictive contribution of these features is persistent across market regimes rather than driven by a few isolated episodes.

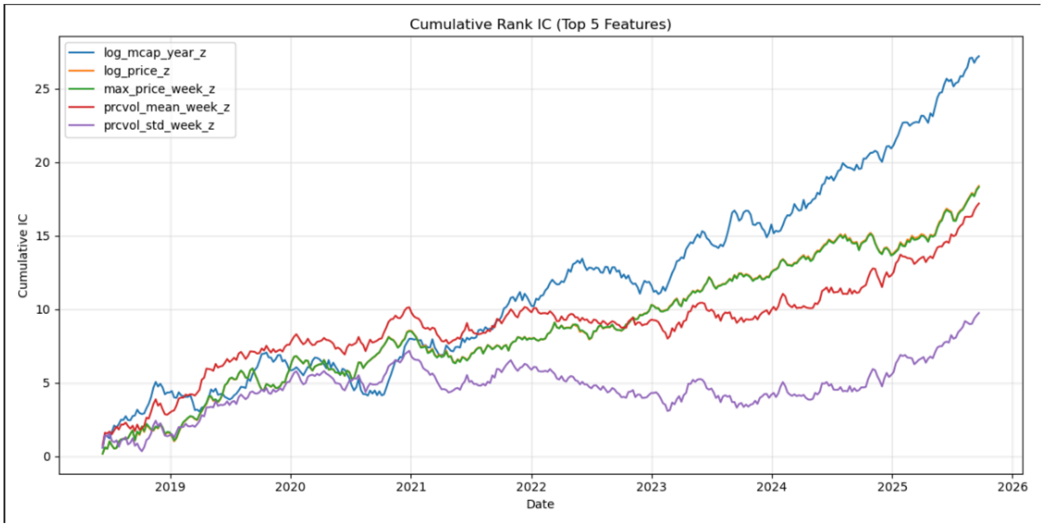

Figure 6: Cumulative RankIC for the five features with the highest positive Mean RankIC.

This evidence supports the interpretation that our cross-sectional ML model systematically rewards large, liquid, and actively traded cryptocurrencies, while penalizing short-term, high-volatility spikes, consistent with a combination of quality/liquidity and mean-reversion effects.

7. Conclusion

This paper proposes and empirically evaluates a weekly cross-sectional cryptocurrency strategy that integrates volatility-scaled machine-learning ranking with beta-neutral portfolio construction and strict risk controls. Using a universe of the top 30 cryptocurrencies by market capitalization, we engineer a comprehensive set of technical, volatility, and liquidity features from weekly OHLCV data and standardize them cross-sectionally. A LightGBM LambdaRank model with DART boosting, combined with a Ridge regression model, is trained on a volatility-scaled target in a rolling walk-forward fashion to generate out-of-sample cross-sectional scores.

An important empirical finding is that the raw ensemble scores are negatively related to future risk-adjusted returns, indicating that the model is particularly effective at identifying over-extended, high-volatility assets. By inverting the signal and implementing it as a beta-neutral, volatility-targeted long–short portfolio, we exploit this mean-reversion behavior and achieve attractive risk-adjusted performance. The strategy delivers a Sharpe ratio around 2 with substantially smaller drawdowns than a Bitcoin buy-and-hold benchmark, despite operating in a highly volatile asset class.

Feature importance and RankIC analyses reveal that the strategy's alpha is driven by two main mechanisms: (i) avoidance or shorting of short-term high-volatility names that tend to mean revert, and (ii) a persistent quality/liquidity tilt toward large, highly traded coins that generate more stable risk-adjusted returns. These effects are intuitive and align with broader empirical findings in both traditional and digital asset markets.

Future work could extend this framework along several dimensions. Enlarging the universe to include mid-cap assets may uncover additional cross-sectional structure but will require careful modeling of liquidity and transaction costs. Incorporating explicit cost-aware objectives inside the ML training loop could further improve implementability. Finally, enriching the feature set with on-chain metrics, funding rates, and derivatives data, as well as considering more advanced architectures such as attention-based temporal models, may enhance predictive power while preserving interpretability through appropriate regularization and diagnostics.

References